

ASSIGNMENT 5

Deadline: 11:59 pm, April 29, 2026

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Please follow the course policy and the school's academic honesty policy.

1 Compulsory Questions

Question 1 (30 %)

Consider the generative classifier where the class conditional densities are multivariate Gaussians:

$$p(x|y = c, \theta) = \mathcal{N}(x|\mu_c, \Sigma_c) = \frac{1}{(2\pi)^{d/2}|\Sigma_c|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_c)^T \Sigma_c^{-1}(x - \mu_c)\right),$$

where d is the dimension of each data x .

Then, the corresponding class posterior has the form:

$$p(y = c|x, \theta) \propto \pi_c \mathcal{N}(x|\mu_c, \Sigma_c),$$

where $\pi_c = p(y = c)$ is the prior probability of class c .

1. Gaussian Discriminant Analysis (GDA) model is the log posterior over class labels: $\log p(y = c|x, \theta)$. Please write the derivation of getting the GDA model's discriminant function.

Hint: You can use "const" to represent the constant in the formula.

2. Only for this sub-question, we consider a 2-classes setting. Please provide the decision boundary equation in terms of $(\pi_c, \mu_c, \Sigma_c, c = 0, 1)$ and explain in which case the boundary will be a linear one.
3. The likelihood function of GDA is as follows:

$$p(\mathcal{D}|\theta) = \prod_{n=1}^N \text{Cat}(y_n|\pi) \prod_{c=1}^C \mathcal{N}(x_n|\mu_c, \Sigma_c)^{\mathbb{I}(y_n=c)},$$

where $\mathcal{D} = \{(x_n, y_n) : n = 1, \dots, N\}$ and $\text{Cat}(y_n|\pi)$ is the categorical distribution (π_c denotes as the probability of $y_n = c$ and N_c denotes as the number of times the event $y_n = c$ is observed). And the indicator function $\mathbb{I}(y_n = c)$ takes the value 1 if $y_n = c$; otherwise, it equals 0.

- (a) Please derive the log-likelihood function $\log p(\mathcal{D}|\theta)$.

Hint: A detailed derivation process is required. A result without the derivation process can gain at most 1 point. You can use the term $\mathcal{N}(x|\mu_c, \Sigma_c)$ to simplify your answer.

- (b) For $p(\mathcal{D}|\theta)$, we have $\theta = \{\pi, \mu_c, \Sigma_c\}$. Please write down the details of how to optimize and get the maximum likelihood estimator $\hat{\theta}_c$.

Hint1: Consider minimizing the negative log likelihood (NLL) subject to the equality constraint with respect to π , using the method of Lagrange multipliers.

Hint2: A detailed derivation process is required. A result without the derivation process can gain at most 1 point.

Question 2 (20 %)

Given the following linearly separable 3D dataset:

- **Class +1:** $(1, 0, 0)$
- **Class -1:** $(1, -\sqrt{2}, 1), (1, \sqrt{2}, 1)$

Your task is to

1. Write the primal hard-margin SVM optimization problem.
2. Derive the Lagrangian and dual problem.
3. Solve the dual analytically for optimal α_i .
4. Recover the primal variables $\mathbf{w}$ and b .
5. State the decision boundary equation.

Question 3 (50 %, Coding Exercise)

In this exercise, you will implement GDA for classification. The training dataset will be provided, containing multiple data samples (x, y) , where x is a feature vector and y is the corresponding class label. You are first required to implement GDA in Python, estimate the model parameters from data, and use the model for classification. After that, you are also required to implement Linear Discriminant Analysis (LDA) in python, which is a special case of GDA. After completing this exercise, you are expected to gain a better understanding of GDA and LDA, as well as the differences between them.

Data Description:

- Training data: `train.csv`
- Test data: `test.csv`
- In the dataset:
 - x is a 2-dimensional feature vector, corresponding to the columns x_1 and x_2 ,
 - y is the class label, taking values in $\{0, 1\}$, corresponding to the column y . Only the training data contains values for y . The test data does not contain the column y and you need to predict the labels for the test data.

Task Description:

1. Use the training data to fit a GDA model via maximum likelihood estimation. You need to estimate the class prior probabilities, the class means, and the class covariance matrices.
2. Use the fitted GDA model to classify the test data and save the classification results.
3. Compute the decision boundary and plot the classification results of the GDA model on the test data. The plot should include all data points and the decision boundary. Different classes of data points should be distinguished using different representations (e.g., different colors).
4. Use the training data to fit an LDA model via MLE. Note that LDA is a special case of GDA in which all classes share the same covariance matrix. Therefore, you can use all data to estimate a shared covariance matrix.
5. Use the fitted LDA model to classify the test data and save the classification results.
6. Compute the decision boundary and plot the classification results of the LDA model on the test data. The requirements for the plot are the same as those for the GDA model.

Submission Requirements:

- A Python file (e.g., `gda.py`) containing your implementation.
- Two prediction result files (one for each model). You should fill in the predicted labels for the test data and save them in two separate files (e.g., `gda_predictions.csv` and `lda_predictions.csv`).
- Two classification plots (one for each model).

Additional Notes:

You may use libraries like `numpy`, `pandas` and `matplotlib` for numerical computations and plotting. Any built-in classification models are forbidden in this exercise (e.g. `LinearDiscriminantAnalysis` in `sklearn(Scikit-learn)`).

2 Optional Questions

Question 4

In a certain town, 60% of taxis are green and 40% are white. Because of poor weather conditions of this town, a witness correctly identifies the color of a taxi with probability 80%. One day, the witness reports seeing a white taxi. What is the probability that the taxi was actually white?

Question 5

Consider a sequence of independent Bernoulli trials, each with the same unknown probability of success probability θ . Let $x_1, x_2, \dots, x_n$ be the observed outcomes, where $x_i = 1$ if the i -th trial is a success, and $x_i = 0$ if it is a failure.

1. Write the maximum likelihood estimate (MLE) of θ based on the observed data $x_1, x_2, \dots, x_n$.
2. Use the MLE to estimate the success rate θ based on the observed data.

Hint: The probability density function of Bernoulli distribution is $f(x) = \theta^x(1-\theta)^{(1-x)}$, where θ is the success rate.

Question 6

The discriminant function for Gaussian Discriminant Analysis (GDA) is given as:

$$\log p(y = c | \mathbf{x}, \theta) = \log \pi_c - \frac{1}{2} \log |2\pi \Sigma_c| - \frac{1}{2} (\mathbf{x} - \mu_c)^T \Sigma_c^{-1} (\mathbf{x} - \mu_c) + \text{const},$$

where π_c is the prior probability of class c , $\mathbf{x}$ denotes the data feature, θ represents the model parameters, μ_c and Σ are the mean and covariance matrix of class c .

1. Please write the decision boundary between two classes c and c' .
2. LDA (Linear Discriminant Analysis) is a special case of GDA where the class covariances are assumed to be equal, i.e.,

$$\Sigma_c = \Sigma, \quad \forall c.$$

Please write the decision boundary between two classes c and c' .

Question 7

We have a dataset for 2 classes.

- For **class 1**, we have these samples:

$$\{(2.1, 1.2, 0.3), (0.9, -0.8, -0.1), (0.2, -1.5, -1.8), (1.5, -0.4, -0.7)\}.$$

- For **class 2**, we have these samples:

$$\{(-0.7, -0.6, -0.9), (-1.0, -1.2, -0.3), (-1.5, 0.4, -1.1), (-0.6, -0.5, 1.0)\}.$$

We want to fit a **Fisher's LDA model** to this dataset. For this question, all results should be rounded to 3 decimal places.

1. Calculate the mean of the two classes, μ_1 and μ_2 .
2. Calculate the between-class scatter matrix $\mathbf{S}_B$.
3. Calculate the within-class scatter matrix $\mathbf{S}_W$.
4. Calculate the optimal projection vector $\mathbf{w}^*$ based on $\mathbf{S}_B$ and $\mathbf{S}_W$.

Homework Submission Policy

- **Full Points:** At most 8 points in total if submitted before the deadline.
- **Late Submission:** At most 6.4 points in total if submitted within 24 hours after the deadline.

- **No Points:** 0 points if submitted more than 24 hours after the deadline.
- You are encouraged to either type your solution in LaTeX or Word, You can also scan or take photos of your handwritten solution and bundle your scans/photos into one single PDF file.
- All written answers should be submitted in a **single PDF file**. Please also compress the **related codes and plots for question 3 into a Zip file** and submit it along with your PDF answer.
- Please submit via Blackboard.